BRAC UNIVERSITY

Principles of Physics-II (PHY-112)

Department of Mathematics and Natural Sciences

Assignment: 02 — **Section**: 02

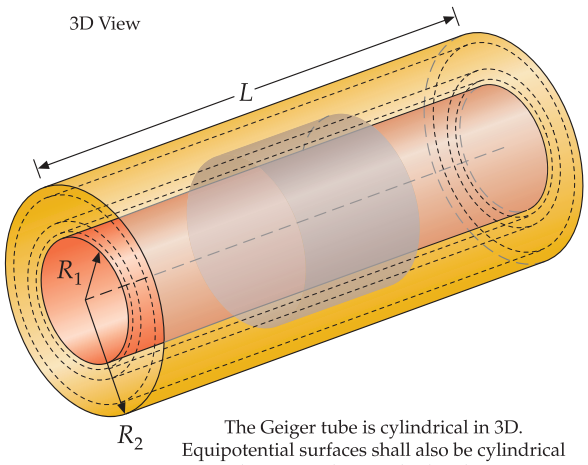
Duration: 7 Days | **Spring 2025 (10F-29C)** | Marks: 30

Assignment Instructions

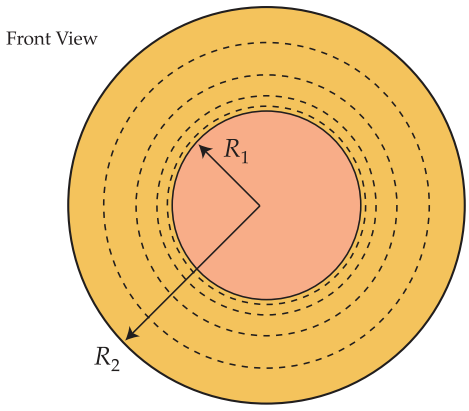
- Write your assignment on A4-sized paper. Use SI units. Properly staple and organize your pages.
- Attempt the questions set for your designated section only. Follow the deadline as written above.
- Ensure your solutions are clear, concise, and well-organized. Add diagrams where needed. Expand your answers when necessary. Do not spit out equations with no context. Steps hold marks.
- **Plagiarism, cheating, copying someone else's script, or the blatant use of AI tools** will result in a 50% decrement or a zero score in extreme cases.
- Late submissions will be penalized at a rate of **10% per day** for two days. No submissions will be accepted two days after the deadline. Especially after the solution is posted. Online submission is not an option.

Alternative solution approaches to the questions are most welcome and will be accepted as long as they do not break the laws of Physics.

1. A Geiger tube comprises two very long, coaxial cylindrical parts. A thin wire is in the center, and a hollow cylinder is around the other. The central wire is positively charged, and the outer cylinder is negatively charged. The potential difference between the wire and the cylinder is 2 kV.
   (a) What is the direction of the electric field inside the tube? Which element is at a higher electric potential? What is (are) the shape(s) of the equipotential surfaces inside the tube? Draw a diagram that shows the cross-section of the tube from the front. **(4)**



The Geiger tube is cylindrical in 3D.
Equipotential surfaces shall also be cylindrical
since they must obey a cylindrical symmetry.



Inner cylinder is positively charged
Outer cylinder is negatively charged
Dashed rings are equipotential surfaces

The electric field in a coaxial cylindrical geometry points **radially outward** from the positively charged central wire toward the negatively charged cylinder.

The central wire is at a higher electric potential because positive charge corresponds to higher potential in electrostatics.

Equipotential surfaces in this cylindrical symmetry are concentric cylinders centered on the wire. They look concentric circular when seen through a cross-section from the front.

(b) Consider two equipotential surfaces described in (a). Suppose they differ in electric potential by 10 V. Do two such equipotential surfaces near the central wire have the same spacing as they would near the outer cylinder? If not, where in the tube are the equipotential surfaces that are more widely spaced? Explain your answer with the same diagram you drew in (a). **(5)**

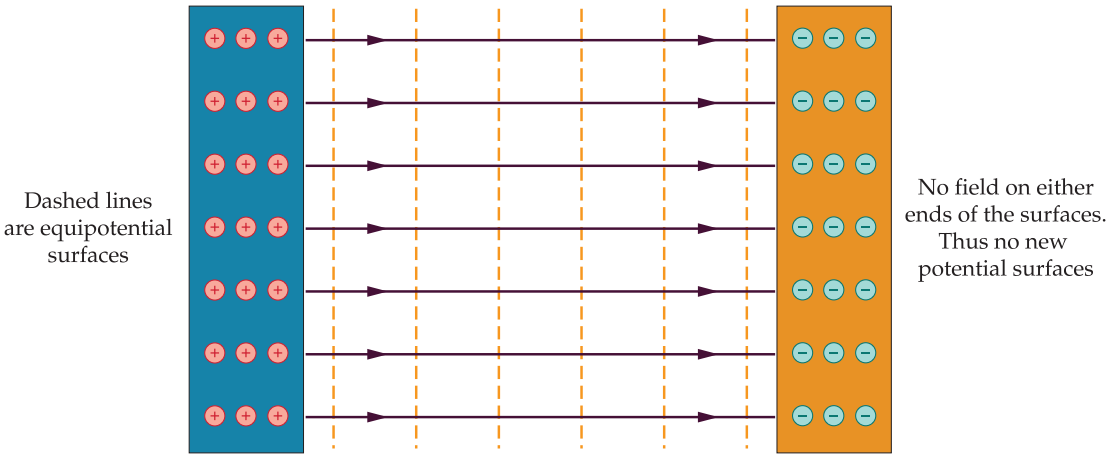
In a cylindrical electric field, the field strength is given by:

$$E(r) = \frac{\lambda}{2\pi\varepsilon_0 r},$$

which shows that the electric field decreases with increasing radius r.

Since $E = -\vec{\nabla}V$, the potential changes more rapidly (i.e., equipotential surfaces are closer together) where the electric field is stronger.

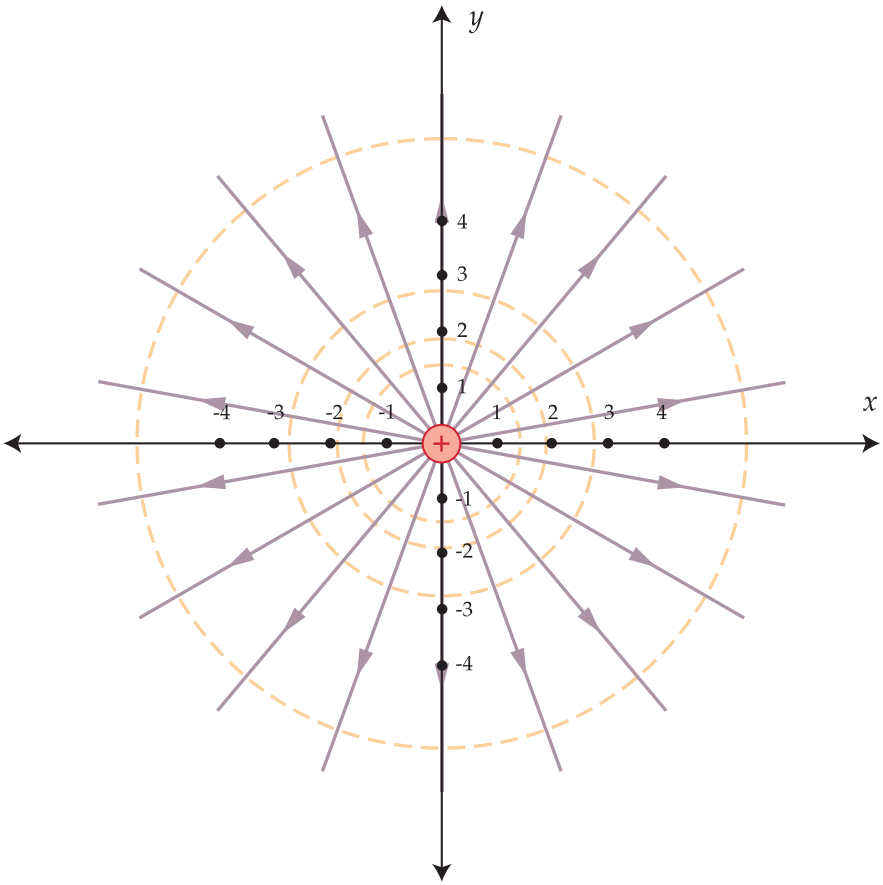
2. Consider two parallel uniformly charged plates that are equal but oppositely charged. (i) What is (are) the shape(s) of the equipotential surfaces in the region between them? Explain your answer. (ii) What is (are) the shape(s) of the equipotential surfaces in the regions on either side of them? Explain your answer with a diagram. **(4)**



Consider two parallel uniformly charged plates with equal and opposite charge.

- The electric field between two large, oppositely charged parallel plates is uniform and directed perpendicular to the plates. In a uniform electric field, equipotential surfaces are always **perpendicular** to the electric field lines.
- Outside the plates, the electric field is no longer uniform and tends to fringe outward. As a result, the equipotential surfaces in these regions are **curved**, forming surfaces that are still perpendicular to the local electric field lines. Thus, the equipotential surfaces are curved and spread outward.

3. The electric field on the x-axis due to a point charge fixed at the origin is given by $\vec{E} = -\frac{a}{y^3}\hat{j}$ where $a = 10\,\text{kV m}$ and $y \neq 0$. (i) Find the magnitude and sign of the point charge. (ii) Find the potential difference between the points on the x-axis at $y = 1.0\,\text{m}$ and $y = 2.0\,\text{m}$. Which of these points has a higher potential? Drawing a diagram is mandatory. **(6)**



(i) The electric field on the y-axis due to a point charge at the origin is given by $\vec{E} = -\frac{a}{y^3}\hat{j}$, where $a = 10\,\text{kV m} = 10^4\,\text{V m}$. This looks like a faulty case where the r dependency does not satisfy a point charge's field condition. We can compare this with the electric field due to a point charge: $E = \frac{C|q|}{y^2}$ as a sanity check.

Equating:

$$\frac{C|q|}{y^2} = \frac{a}{y^3} \Rightarrow C|q| = \frac{a}{y} \Rightarrow |q| = \frac{a}{Cy} = \frac{10^4}{8.988 \times 10^9 y} = \frac{1.112}{y}\,\text{nC}.$$

As expected, we see an impossible dimension for point charge above which claims charge depends on location y. No such point charge can exist as the question claims. This part of the question cannot have a valid answer for a point charge or its magnitude. The field however looks more that of an electric dipole's along its dipole axis.

(ii) To find the potential difference between $y = 1.0\,\text{m}$ and $y = 2.0\,\text{m}$ (assuming it's a dipole source):

$$\Delta V = -\int_1^2 \vec{E}\cdot d\vec{l} = -\int_1^2 \frac{-a}{y^3}dy = a\left[\frac{1}{2y^2}\right]_1^2$$
$$= a\left(\frac{1}{8} - \frac{1}{2}\right) = -a\left(\frac{3}{8}\right) = \frac{10^4 \cdot 3}{4} = -3750\,\text{V}.$$

So, $V(2\,\text{m}) - V(1\,\text{m}) = -3750\,\text{V} \Rightarrow V(1\,\text{m}) > V(2\,\text{m})$.

4. One non-conducting rubber ball and one conducting metal ball, each of radius 2.5 cm, are placed 1.5 m apart from center to center. Ball 1 has a charge of $+2.0 \times 10^{-8}\,\text{C}$, and ball 2 has a charge of $-1.0 \times 10^{-8}\,\text{C}$. The balls are far enough apart that they don't affect each other's charge distribution. Take the electric potential to be zero at infinity. (i) What is the electric potential at the point exactly halfway between the centers of the two balls? (ii) What is the net potential 0.5 cm inside the surface of (ii) ball 1 and (iii) ball 2, respectively, one at a time? **(6)**

Two nonconducting balls of radius $R = 2.5\,\text{cm} = 0.025\,\text{m}$ are placed $d = 1.5\,\text{m}$ apart.
Ball 1: $q_1 = +2.0 \times 10^{-8}\,\text{C}$, Ball 2: $q_2 = -1.0 \times 10^{-8}\,\text{C}$.

The electric potential due to a point or spherically symmetric charge distribution is $V = \dfrac{kq}{r}$ for points outside the charge.

The point halfway between the centers is at $r_1 = r_2 = \dfrac{1.5}{2} = 0.75\,\text{m}$ from each center.
The total potential at this point is the algebraic sum:

$$\begin{aligned} V &= \frac{Cq_1}{r_1} + \frac{Cq_2}{r_2} = \frac{(8.988 \times 10^9)(2.0 \times 10^{-8})}{0.75} + \frac{(8.988 \times 10^9)(-1.0 \times 10^{-8})}{0.75} \\ &= 239.68\,\text{V} - 119.84\,\text{V} = 119.84\,\text{V}. \end{aligned}$$

Potential just inside the surface of ball 1 (nonconducting) ($r = R = 0.025\,\text{m}$):

$$\begin{aligned} V_1 &= \frac{q_1(3R_1^2 - r^2)}{8\pi\epsilon_0 R_1^3} \\ &= \frac{(8.988 \times 10^9)\,(2 \times 10^{-8})\,\left[3\,(2.5 \times 10^{-2})^2 - (0.5 \times 10^{-2})^2\right]}{2 \times (2.5 \times 10^{-2})^3} = 10641.792\,\text{V}. \end{aligned}$$

Potential due to ball 2 at that location:

$$V_2 = \frac{Cq_2}{R_2} = \frac{(8.988 \times 10^9)(-1.0 \times 10^{-8})}{(1.5 + 2.5 \times 10^{-2})} = -58.938\,\text{V}.$$

Net potential in this case:

$$\begin{aligned} V_{\text{net}} &= V_1 + V_2 \\ &= 1064.1792\,\text{V} - 5.8938\,\text{V} \\ &= 10582.854\,\text{V} \end{aligned}$$

Repeat the calculation for measurement inside the conducting ball: Potential just inside the surface of ball 2 (conducting) is identical to that of the surface:

$$\begin{aligned} V_2 &= \frac{Cq_2}{R_2} \\ &= \frac{(8.988 \times 10^9)\,(-10^{-8})}{(2.5 \times 10^{-2})} = -3595.2\,\text{V}. \end{aligned}$$

Potential due to ball 2 at that location:

$$V_1 = \frac{Cq_1}{R_1} = \frac{(8.988 \times 10^9)(2 \times 10^{-8})}{(1.5 + 2.5 \times 10^{-2})} = 117.875\,\text{V}.$$

Net potential in this case:

$$\begin{aligned} V_{\text{net}} &= V_1 + V_2 \\ &= -3595.2\,\text{V} + 117.875\,\text{V} \\ &= -3477.32459\,\text{V} \end{aligned}$$

5. While building an electric scooter, you need to temporarily splice, end to end, a 1.2 m iron wire with an 8.0 m nickel wire that has the same diameter. The maximum current in the wires is 1.5 A. (i) Find the (i) resistances and (ii) potential drop across each wire of this system when the current is 1.5 A. (iii) Find the electric field in each wire. Where does this electric field come from? Is it uniform or non-uniform? **(5)**

The potential drop across a wire is given by $\Delta V = IR$, and the electric field by $E = \dfrac{\Delta V}{L}$.

Resistivity values: $\rho_{\text{Fe}} = 9.579 \times 10^{-8}\,\Omega\cdot$, $\rho_{\text{Ni}} = 6.99 \times 10^{-8}\,\Omega\cdot$. (Google them)

Let the cross-sectional area be A, same for both wires.

$$R_{\mathrm{Fe}} = \frac{\rho_{\mathrm{Fe}} L_{\mathrm{Fe}}}{A} = \frac{(9.579 \times 10^{-8})(1.2)}{A} = \frac{1.14948 \times 10^{-7}}{A}\,\Omega,$$
$$R_{\mathrm{Ni}} = \frac{\rho_{\mathrm{Ni}} L_{\mathrm{Ni}}}{A} = \frac{(6.99 \times 10^{-8})(8.0)}{A} = \frac{5.592 \times 10^{-7}}{A}\,\Omega.$$

$$V_{\mathrm{Fe}} = IR_{\mathrm{Fe}} = (1.5) \cdot \frac{1.14948 \times 10^{-7}}{A} = \frac{1.72422 \times 10^{-7}}{A}\,\mathrm{V},$$
$$V_{\mathrm{Ni}} = IR_{\mathrm{Ni}} = (1.5) \cdot \frac{5.592 \times 10^{-7}}{A} = \frac{8.388 \times 10^{-7}}{A}\,\mathrm{V}.$$

$$E_{\mathrm{Fe}} = \frac{\Delta V_{\mathrm{Fe}}}{L_{\mathrm{Fe}}} = \frac{1.7478 \times 10^{-7}}{A \cdot 1.2} = \frac{1.4565 \times 10^{-7}}{A}\,\mathrm{V\,m^{-1}},$$
$$E_{\mathrm{Ni}} = \frac{\Delta V_{\mathrm{Ni}}}{L_{\mathrm{Ni}}} = \frac{8.388 \times 10^{-7}}{A \cdot 8.0} = \frac{1.0485 \times 10^{-7}}{A}\,\mathrm{V\,m^{-1}}.$$